A1.1 Water

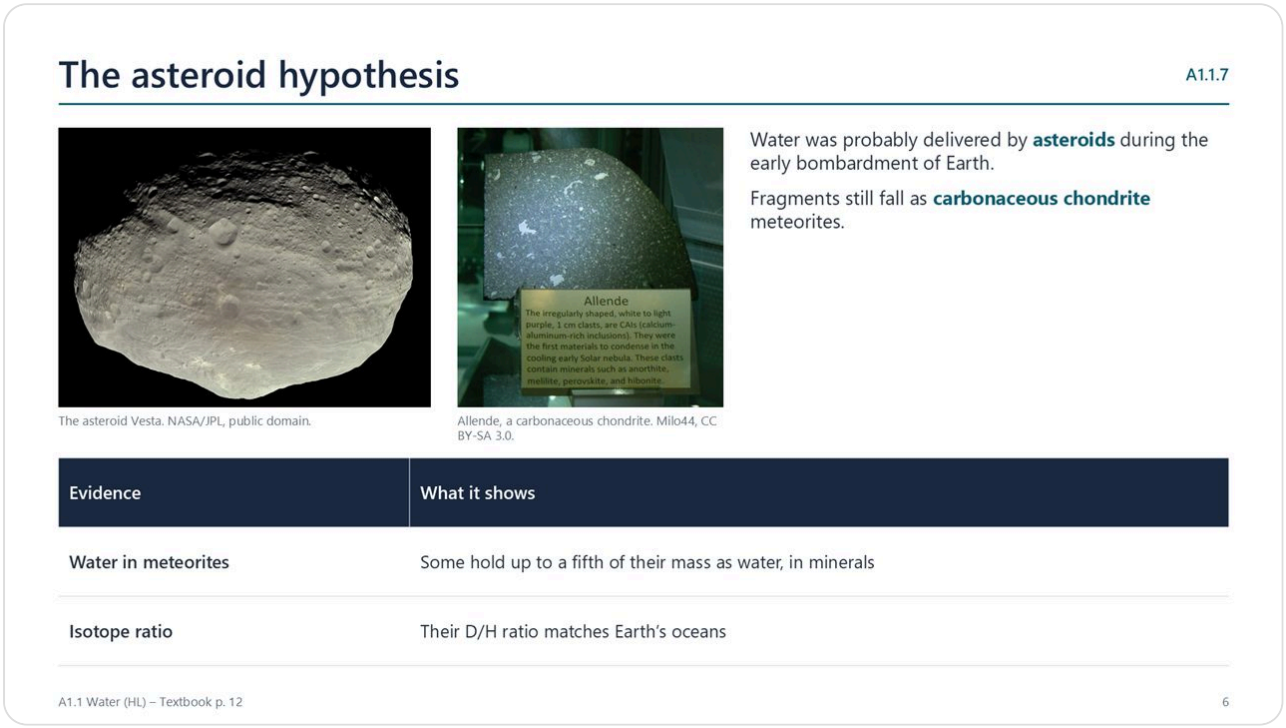
Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-a1-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 11–13 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Water on Earth and beyond

A1.1.7 Where Earth's water came from

pp. 11–12



The Earth formed about 4.5 billion years ago, too close to the Sun and too hot for water vapour to condense. Water ice could form only further out, so most of Earth's water is thought to be **extraplanetary**, delivered by **asteroids**. Some carbonaceous chondrite meteorites hold water in their minerals, with a **deuterium-to-hydrogen ratio** close to that of seawater. Earth kept its water because its **gravity** holds water vapour and an atmosphere, and it is **cool enough** for vapour to condense into oceans. Smaller Mars lost most of its atmosphere and surface water.

A1.1.8 Water and the search for life

p. 13

All known life needs liquid water as a solvent, so the search for life elsewhere begins with water. The **Goldilocks (habitable) zone** is the range of distances from a star where a planet's surface could hold liquid water. Astronomers detect water in the atmospheres of distant planets by **transit spectroscopy**. Being in the zone does not guarantee life: a planet also needs enough mass, an atmosphere and time. Moons such as Europa may hold oceans under ice even outside the zone.

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